

SAFETY DATA SHEET

THYPAAN VERTEX



Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	26.11.2025	50001099	Date of first issue: 26.11.2025

1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product name : THYPAAN VERTEX

Manufacturer or supplier's details

Company : FMC Agro Kazakhstan LLP

Address : Timiryazev str. 28V, office 501, 5 floor
050040 Almaty
Kazakhstan

Emergency telephone number : +44 20 3885 0382 (CHEMTREC's European Regional Toll-Free Number)
1 703 / 741-5970 (CHEMTREC - International)
1 703 / 527-3887 (CHEMTREC - Alternate)

Medical Emergency Number : Ambulance in the Republic of Kazakhstan: 103 (24/7, toll-free).
Almaty City Clinical Hospital, Toxicology Department, 96 Kazybek Bi Street: +7 727 317 01 69 – hotline; +7 727 292 31 01 – reception.
+1 651 / 632-6793 (outside USA & Canada, collect call)

E-mail address : SDS-Info@fmc.com

Recommended use of the chemical and restrictions on use

Recommended use : Crop nutrition

Restrictions on use : Use as recommended by the label.

2. HAZARDS IDENTIFICATION

GHS Classification

Skin irritation : Category 2

Serious eye damage : Category 1

Short-term (acute) aquatic hazard : Category 2

Long-term (chronic) aquatic hazard : Category 2

GHS-Labeling

Hazard pictograms :



Signal word : Danger

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Hazard statements : H315 Causes skin irritation.
H318 Causes serious eye damage.
H411 Toxic to aquatic life with long lasting effects.

Precautionary statements : **Prevention:**
P220 Keep away from clothing and other combustible materials.
P264 Wash skin thoroughly after handling.
P273 Avoid release to the environment.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.
Response:
P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/ doctor.
Disposal:
P501 Dispose of contents/ container to an approved facility in accordance with local, regional, national and international regulations.

Other hazards which do not result in classification

None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Pure substance/mixture : Mixture

Components

Chemical name	CAS-No.	Classification	MAC value mg/m ³ / TSEL value	Concentration (% w/w)
ammonium nitrate	6484-52-2	Ox. Sol.3; H272 Acute Tox.5; H303 Eye Irrit.2A; H319	No data available	>= 20 - < 30
urea	57-13-6	No data available	MPC-STEL: 10 mg/m ³ Class 3 - Moder- ately dangerous Data Source: RU OEL	>= 20 - < 30
magnesium nitrate	10377-60-3	Ox. Sol.3; H272 Acute Tox.5; H303 Eye Irrit.2A; H319	No data available	>= 1 - < 10

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manganese dinitrate	10377-66-9	Ox. Sol.3; H272 Acute Tox.4; H302 Skin Corr.1C; H314 Eye Dam.1; H318 STOT RE2; H373 Aquatic Acute3; H402 Aquatic Chronic1; H410	No data available	>= 1 - < 2.5
copper dinitrate	3251-23-8	Ox. Sol.2; H272 Skin Corr.1B; H314 Eye Dam.1; H318 Aquatic Acute1; H400 Aquatic Chronic1; H410	No data available	>= 1 - < 2.5
nitric acid ...% [C ≤ 70 %]	7697-37-2	Ox. Liq.2; H272 Met. Corr.1; H290 Acute Tox.3; H331 Skin Corr.1A; H314 Eye Dam.1; H318 Muta.2; H341	No data available	>= 0.1 - < 1

For explanation of abbreviations see section 16.

4. FIRST AID MEASURES

- General advice : Move out of dangerous area.
Consult a physician.
Show this safety data sheet to the doctor in attendance.
Do not leave the victim unattended.
- If inhaled : Move to fresh air.
If unconscious, place in recovery position and seek medical advice.
If symptoms persist, call a physician.
If experiencing any discomfort, immediately remove from exposure. Light cases: Keep person under surveillance. Get medical attention immediately if symptoms develop. Serious cases: Get medical attention immediately or call for an ambulance.

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- lance.
- In case of skin contact : Take off all contaminated clothing immediately.
Wash contaminated clothing before re-use.
Wash off immediately with plenty of water for at least 15 minutes.
Get medical attention if irritation develops and persists.
- In case of eye contact : Small amounts splashed into eyes can cause irreversible tissue damage and blindness.
In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
Continue rinsing eyes during transport to hospital.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.
- If swallowed : Do not induce vomiting without medical advice.
Keep respiratory tract clear.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.
- Most important symptoms and effects, both acute and delayed : Causes skin irritation.
Causes serious eye damage.
- Protection of first-aiders : First Aid responders should pay attention to self-protection and use the recommended protective clothing
Avoid inhalation, ingestion and contact with skin and eyes.
If potential for exposure exists refer to Section 8 for specific personal protective equipment.
- Notes to physician : Treat symptomatically.

5. FIREFIGHTING MEASURES

- Suitable extinguishing media : Dry chemical, CO₂, water spray or regular foam.
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
- Unsuitable extinguishing media : High volume water jet
Do not spread spilled material with high-pressure water streams.
- Specific hazards during fire-fighting : Do not allow run-off from fire fighting to enter drains or water courses.
- Hazardous combustion products : Ammonia
Fire may produce irritating, corrosive and/or toxic gases.
- Specific extinguishing methods : Collect contaminated fire extinguishing water separately. This must not be discharged into drains.

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Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.
For safety reasons in case of fire, cans should be stored separately in closed containments.
Use a water spray to cool fully closed containers.

Special protective equipment : Firefighters should wear protective clothing and self-contained breathing apparatus.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures : Use personal protective equipment.
Ensure adequate ventilation.
If it can be safely done, stop the leak.
Do not touch or walk through the spilled material.
Never return spills in original containers for re-use.
Mark the contaminated area with signs and prevent access to unauthorized personnel.
Only qualified personnel equipped with suitable protective equipment may intervene.
For disposal considerations see section 13.

Environmental precautions : Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.
If the product contaminates rivers and lakes or drains inform respective authorities.

Methods and materials for containment and cleaning up : Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).

7. HANDLING AND STORAGE

Advice on protection against fire and explosion : Keep away from combustible material.

Advice on safe handling : Do not breathe vapours/dust.
Avoid exposure - obtain special instructions before use.
Avoid contact with skin and eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
To avoid spills during handling keep bottle on a metal tray.
Dispose of rinse water in accordance with local and national regulations.

Conditions for safe storage : Keep container tightly closed in a dry and well-ventilated place.
Containers which are opened must be carefully resealed and kept upright to prevent leakage.
Electrical installations / working materials must comply with the technological safety standards.

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Materials to avoid : Do not store near acids.

Further information on storage stability : No decomposition if stored and applied as directed.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Personal protective equipment

Respiratory protection : No personal respiratory protective equipment normally required.

Hand protection
Material : Wear chemical resistant gloves, such as barrier laminate, butyl rubber or nitrile rubber.

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Eye protection : Eye wash bottle with pure water
Tightly fitting safety goggles
Wear face-shield and protective suit for abnormal processing problems.

Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Protective measures : Wear suitable protective equipment.
Plan first aid action before beginning work with this product.
Always have on hand a first-aid kit, together with proper instructions.

Hygiene measures : When using do not eat or drink.
When using do not smoke.
Wash hands before breaks and at the end of workday.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state : liquid

Form : liquid

Colour : blue green

Odour : Faint odour

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Odour Threshold	:	No data available
pH	:	3.0 - 4.5 Concentration: 100 %
Melting point/freezing point	:	No data available
Initial boiling point and boiling range	:	No data available
Flash point	:	No data available
Upper explosion limit / Upper flammability limit	:	No data available
Lower explosion limit / Lower flammability limit	:	No data available
Vapour pressure	:	No data available
Relative vapour density	:	No data available
Relative density	:	1.33 - 1.35
Density	:	No data available
Bulk density	:	No data available
Solubility(ies)		
Water solubility	:	soluble
Solubility in other solvents	:	No data available
Partition coefficient: n-octanol/water	:	No data available
Auto-ignition temperature	:	No data available
Decomposition temperature	:	No data available
Viscosity		
Viscosity, dynamic	:	No data available
Viscosity, kinematic	:	No data available
Explosive properties	:	No data available
Oxidizing properties	:	Non-oxidizing
Particle size	:	No data available

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10. STABILITY AND REACTIVITY

Reactivity	: No decomposition if stored and applied as directed.
Chemical stability	: No decomposition if stored and applied as directed.
Possibility of hazardous reactions	: No decomposition if stored and applied as directed.
Conditions to avoid	: Avoid extreme temperatures Avoid formation of aerosol.
Incompatible materials	: Avoid strong acids, bases, and oxidizers
Hazardous decomposition products	: No decomposition if stored and applied as directed.

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Based on available data, the classification criteria are not met.

Product:

Acute oral toxicity	: Acute toxicity estimate: > 5,000 mg/kg Method: Calculation method
Acute inhalation toxicity	: Acute toxicity estimate: > 40 mg/l Exposure time: 4 h Test atmosphere: vapour Method: Calculation method
Acute dermal toxicity	: Remarks: No data is available on the product itself.

Components:

water:

Acute oral toxicity	: Remarks: No data available
Acute inhalation toxicity	: Remarks: No data available

ammonium nitrate:

Acute oral toxicity	: LD50 (Rat, male and female): 2,950 mg/kg Method: OECD Test Guideline 401
Acute dermal toxicity	: LD50 (Rat, male and female): > 5,000 mg/kg Method: OECD Test Guideline 402

urea:

Acute oral toxicity	: LD50 (Rat, male and female): > 5,000 mg/kg
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magnesium nitrate:

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Acute oral toxicity : LD50 (Rat, female): > 2,000 mg/kg
Method: OECD Test Guideline 423

Acute dermal toxicity : LD50 (Rat, male and female): > 5,000 mg/kg
Method: OECD Test Guideline 402

manganese dinitrate:

Acute oral toxicity : LD50 Oral (Rat, female): > 300 mg/kg
Method: OECD Test Guideline 420

nitric acid ...% [C ≤ 70 %]:

Acute inhalation toxicity : LC50 (Rat, male and female): > 2.65 mg/l
Exposure time: 4 h
Test atmosphere: vapour
Method: OECD Test Guideline 403
Assessment: Corrosive to the respiratory tract.

Skin corrosion/irritation

Causes skin irritation.

Product:

Assessment : Irritating to skin.
Result : Skin irritation

Components:

ammonium nitrate:

Species : Rabbit
Method : OECD Test Guideline 404
Result : No skin irritation

urea:

Species : Rabbit
Method : OECD Test Guideline 404
Result : No skin irritation

magnesium nitrate:

Species : Rabbit
Method : OECD Test Guideline 404
Result : No skin irritation
Remarks : Based on data from similar materials

manganese dinitrate:

Species : Rabbit
Method : OECD Test Guideline 404
Result : Corrosive after 1 to 4 hours of exposure

copper dinitrate:

Method : OECD Test Guideline 431

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Result : Corrosive after 3 minutes to 1 hour of exposure

nitric acid ...% [C ≤ 70 %]:

Result : Corrosive after 3 minutes or less of exposure

Serious eye damage/eye irritation

Causes serious eye damage.

Product:

Result : irritating
Assessment : Irritating to eyes.

Components:

ammonium nitrate:

Species : Rabbit
Result : Irritation to eyes, reversing within 21 days
Method : OECD Test Guideline 405

urea:

Species : Rabbit
Result : No eye irritation
Method : OECD Test Guideline 405

magnesium nitrate:

Species : Rabbit
Result : Eye irritation
Method : OECD Test Guideline 405

manganese dinitrate:

Species : Bovine cornea
Result : Irreversible effects on the eye

Respiratory or skin sensitisation

Skin sensitisation

Based on available data, the classification criteria are not met.

Respiratory sensitisation

Based on available data, the classification criteria are not met.

Product:

Remarks : No data is available on the product itself.

Components:

ammonium nitrate:

Test Type : Local lymph node assay (LLNA)
Species : Mouse
Method : OECD Test Guideline 429

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Result : Does not cause skin sensitisation.

magnesium nitrate:

Test Type : Local lymph node assay (LLNA)
Species : Mouse
Method : OECD Test Guideline 429
Result : Does not cause skin sensitisation.

manganese dinitrate:

Test Type : Local lymph node assay (LLNA)
Species : Mouse
Method : OECD Test Guideline 429
Result : Does not cause skin sensitisation.

copper dinitrate:

Test Type : Maximisation Test
Exposure routes : Dermal
Species : Guinea pig
Method : OECD Test Guideline 406
Result : Does not cause skin sensitisation.

Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Components:**ammonium nitrate:**

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: OECD Test Guideline 471
Result: negative
Remarks: Based on data from similar materials

Test Type: In vitro mammalian cell gene mutation test
Method: OECD Test Guideline 476
Result: negative
Remarks: Based on data from similar materials

Test Type: Chromosome aberration test in vitro
Method: OECD Test Guideline 473
Result: negative

Germ cell mutagenicity - Assessment : In vitro tests did not show mutagenic effects

urea:

Genotoxicity in vitro : Test Type: reverse mutation assay
Result: negative

magnesium nitrate:

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: OECD Test Guideline 471

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Result: negative

Test Type: Chromosome aberration test in vitro
Method: OECD Test Guideline 473
Result: negative
Remarks: Based on data from similar materials

Test Type: In vitro mammalian cell gene mutation test
Method: OECD Test Guideline 476
Result: negative
Remarks: Based on data from similar materials

Germ cell mutagenicity - Assessment : In vitro tests did not show mutagenic effects

manganese dinitrate:

Genotoxicity in vitro : Test Type: In vitro mammalian cell gene mutation test
Method: OECD Test Guideline 476
Result: negative

Test Type: reverse mutation assay
Method: OECD Test Guideline 471
Result: negative

Test Type: Chromosome aberration test in vitro
Method: OECD Test Guideline 473
Result: negative

Genotoxicity in vivo : Test Type: In vivo micronucleus test
Species: Mouse (female)
Application Route: Oral
Method: OECD Test Guideline 474
Result: negative

Germ cell mutagenicity - Assessment : Weight of evidence does not support classification as a germ cell mutagen.

copper dinitrate:

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: OECD Test Guideline 471
Result: negative

Genotoxicity in vivo : Test Type: unscheduled DNA synthesis assay
Species: Rat (male)
Application Route: Oral
Method: OECD Test Guideline 486
Result: negative

Test Type: Micronucleus test
Species: Mouse (male and female)
Application Route: Oral
Method: Mutagenicity (micronucleus test)
Result: negative

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nitric acid ...% [C ≤ 70 %]:

Genotoxicity in vitro : Test Type: reverse mutation assay
Result: negative

Test Type: Chromosome aberration test in vitro
Method: OECD Test Guideline 473
Result: negative
Remarks: Based on data from similar materials

Test Type: Chromosome aberration test in vitro
Result: positive
Remarks: Based on data from similar materials

Test Type: In vitro mammalian cell gene mutation test
Method: OECD Test Guideline 476
Result: negative
Remarks: Based on data from similar materials

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse (male)
Application Route: Oral
Result: positive
Remarks: Based on data from similar materials

Test Type: unscheduled DNA synthesis assay
Species: Mouse (male)
Application Route: Oral
Result: negative
Remarks: Based on data from similar materials

Germ cell mutagenicity - Assessment : Positive result(s) from in vivo somatic cell mutagenicity tests supported by positive results from in vitro mutagenicity assays or chemical structure activity relationship to known germ cell mutagens

Carcinogenicity

Based on available data, the classification criteria are not met.

Components:**urea:**

Species : Rat
Application Route : Oral
Exposure time : 12 month(s)
Result : negative

manganese dinitrate:

Species : Rat, male
Application Route : Oral
Exposure time : 103 weeks
Dose : 60, 200, 615 mg/kg body weight
: 615 mg/kg body weight
Result : negative

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Carcinogenicity - Assessment : Weight of evidence does not support classification as a carcinogen

Reproductive toxicity

Based on available data, the classification criteria are not met.

Components:

ammonium nitrate:

Effects on fertility : Species: Rat, male and female
Application Route: Oral
Dose: 0, 250, 750, and 1,500 milligram per kilogram
General Toxicity - Parent: NOAEL: $\geq$ 1,500 mg/kg body weight
Method: OECD Test Guideline 422
Result: negative
Remarks: Based on data from similar materials

Effects on foetal development : Species: Rat, male and female
Application Route: Oral
Dose: 0, 250, 750, and 1,500 milligram per kilogram
General Toxicity Maternal: NOAEL: $\geq$ 1,500 mg/kg body weight
Developmental Toxicity: NOAEL: $\geq$ 1,500 mg/kg body weight
Method: OECD Test Guideline 422
Result: negative
Remarks: Based on data from similar materials

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

urea:

Effects on foetal development : Test Type: Embryo-foetal development
Species: Rat
Application Route: Oral
Method: OECD Test Guideline 414
Result: negative

magnesium nitrate:

Effects on fertility : Species: Rat, male and female
Application Route: Oral
Dose: 0, 250, 750, and 1,500 milligram per kilogram
Duration of Single Treatment: 28 d
General Toxicity - Parent: NOAEL: $>$ 1,500 mg/kg body weight
Method: OECD Test Guideline 422
Result: negative
Remarks: Based on data from similar materials

Effects on foetal development : Species: Rat
Application Route: Oral
Dose: 0, 250, 750, and 1,500 milligram per kilogram
Duration of Single Treatment: 28 d
General Toxicity Maternal: NOAEL: $>$ 1,500 mg/kg body

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weight
Developmental Toxicity: NOAEL: > 1,500 mg/kg body weight
Method: OECD Test Guideline 422
Result: negative
Remarks: Based on data from similar materials

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

manganese dinitrate:

Effects on fertility : Test Type: Two-generation study
Species: Rat, male and female
Application Route: inhalation (dust/mist/fume)
Dose: 0, 5, 10, 20 µg/L
General Toxicity - Parent: NOEC: 0.020 mg/l
General Toxicity F1: NOAEC: 0.020 mg/l
Method: OECD Test Guideline 416
Result: negative

Effects on foetal development : Species: Rat
Application Route: inhalation (dust/mist/fume)
General Toxicity Maternal: NOAEL: 0.005 mg/L
Embryo-foetal toxicity: NOAEL: 0.015 mg/L
Method: OECD Test Guideline 414

nitric acid ...% [C ≤ 70 %]:

Effects on fertility : Test Type: reproductive and developmental toxicity study
Species: Rat, male and female
Application Route: Oral
Dose: 0, 270, 750, 1500 mg/kg/day
General Toxicity - Parent: NOAEL: 1,500 mg/kg bw/day
General Toxicity F1: NOAEL: ≥ 1,500 mg/kg bw/day
Method: OECD Test Guideline 422
Result: negative
Remarks: Based on data from similar materials

Effects on foetal development : Test Type: reproductive and developmental toxicity study
Species: Rat
Application Route: Oral
Duration of Single Treatment: 53 d
General Toxicity Maternal: NOAEL: 1,500 mg/kg bw/day
Embryo-foetal toxicity: NOAEL: 1,500 mg/kg bw/day
Method: OECD Test Guideline 422
Result: negative
Remarks: Based on data from similar materials

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

STOT - single exposure

Based on available data, the classification criteria are not met.

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Components:

nitric acid ...% [C ≤ 70 %]:

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

STOT - repeated exposure

Based on available data, the classification criteria are not met.

Components:

ammonium nitrate:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

magnesium nitrate:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

manganese dinitrate:

Assessment : The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 2.

Repeated dose toxicity

Components:

ammonium nitrate:

Species : Rat, male
NOAEL : 256 mg/kg
Application Route : Oral
Exposure time : 1 year
Dose : 42, 256, 1527 mg/kg bw/day
Method : OECD Test Guideline 453
Symptoms : No adverse effects
Remarks : Based on data from similar materials

Species : Rat, female
NOAEL : 284 mg/kg
Application Route : Oral
Exposure time : 1 year
Dose : 48, 284, 1490 mg/kg bw/d
Method : OECD Test Guideline 453
Symptoms : No adverse effects
Remarks : Based on data from similar materials

Species : Guinea pig, male
NOAEC : 0.001 mg/l
Application Route : Inhalation
Exposure time : 4 weeks
Dose : 1 mg/m³
Method : OECD Test Guideline 412
Symptoms : No adverse effects

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Species	: Rat, male
NOAEC	: 0.001 mg/l
Application Route	: Inhalation
Exposure time	: 4 weeks
Dose	: 1 mg/m ³
Method	: OECD Test Guideline 412
Symptoms	: No adverse effects

urea:

Species	: Mouse
NOAEL	: 45,000 mg/kg
Application Route	: Oral
Exposure time	: 12 months

magnesium nitrate:

Species	: Rat, male and female
NOAEL	: > 1,500 mg/kg
Application Route	: Oral
Exposure time	: 28d
Dose	: 0, 250, 750, 1,500 mg/kg/day
Method	: OECD Test Guideline 422
Remarks	: Based on data from similar materials

manganese dinitrate:

Species	: Rat, male
NOAEL	: 1700 mg/kg bw/day
Application Route	: Oral
Exposure time	: 13weeks
Dose	: 110 to 1700 mg/kg

Species	: Rat, male and female
NOAEL	: 20 µg/L air
Application Route	: inhalation (dust/mist/fume)
Dose	: 5, 10, 20 µg/L air
Method	: OPPTS 870.3800

nitric acid ...% [C ≤ 70 %]:

Species	: Rat, male and female
NOAEL	: 1500 mg/kg bw/day
Application Route	: Oral
Dose	: 0, 250, 750, 1500 mg/kg/day
Method	: OECD Test Guideline 422
Remarks	: Based on data from similar materials

Species	: Rabbit, male
LOEC	: 50 µg/m ³
Application Route	: Inhalation
Test atmosphere	: vapour
Exposure time	: 4 weeks
Dose	: 50, 150, and 450 µg/m ³

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Aspiration toxicity

Based on available data, the classification criteria are not met.

Further information

Product:

Remarks : No data available

12. ECOLOGICAL INFORMATION

Ecotoxicity

Components:

water:

Toxicity to fish : Remarks: No data available

ammonium nitrate:

Toxicity to fish : LC50 (Cyprinus carpio (Carp)): 95 - 102 mg/l
Exposure time: 48 h
Test Type: semi-static test

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 490 mg/l
Exposure time: 48 h
Remarks: Based on data from similar materials

Toxicity to algae/aquatic plants : EC50 (Marine Diatom): > 1,700 mg/l
Exposure time: 10 d
Test Type: static test
Remarks: Based on data from similar materials

Toxicity to microorganisms : EC50 (activated sludge): > 1,000 mg/l
Exposure time: 3 h
Method: OECD Test Guideline 209
Remarks: Based on data from similar materials

urea:

Toxicity to fish : LC50 (Leuciscus idus (Golden orfe)): 6,810 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia (water flea)): 10,000 mg/l
Exposure time: 48 h

Toxicity to algae/aquatic plants : NOEC (blue-green algae): 47 mg/l
Exposure time: 72 h

Toxicity to microorganisms : No data available (Pseudomonas putida): 10,000 mg/l
Exposure time: 16 h

magnesium nitrate:

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): > 100 mg/l
Exposure time: 96 h

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Method: OECD Test Guideline 203
Remarks: Based on data from similar materials

LC50 (Poecilia reticulata (guppy)): 1,378 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203
Remarks: Based on data from similar materials

LC50 (Cyprinus carpio (Carp)): 95 - 102 mg/l
Exposure time: 48 h
Test Type: semi-static test
Remarks: Based on data from similar materials

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 39 mg/l
Exposure time: 96 h
Remarks: Based on data from similar materials

Toxicity to algae/aquatic plants : EC50 (diatoms): > 1,700 mg/l
Exposure time: 10 d
Test Type: static test
Remarks: Based on data from similar materials

Toxicity to fish (Chronic toxicity) : NOEC (Pimephales promelas (fathead minnow)): 58 mg/l
Exposure time: 30 d
Test Type: flow-through test
Remarks: Based on data from similar materials

NOEC (Pimephales promelas (fathead minnow)): 157 mg/l
Exposure time: 32 d
Test Type: flow-through test
Remarks: Based on data from similar materials

Toxicity to microorganisms : EC50 (activated sludge): > 1,000 mg/l
Exposure time: 3 h
Method: OECD Test Guideline 209
Remarks: Based on data from similar materials

manganese dinitrate:

Toxicity to fish : LC50 (Fish): 55.26 - 67.71 mg/l
Exposure time: 96 h
Test Type: static test

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 100 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : LOEC (Lemna minor (duckweed)): 64.94 mg/l
Exposure time: 7 d
Method: OECD Test Guideline 221
Remarks: Based on data from similar materials

EC10 (Lemna minor (duckweed)): 23.37 mg/l
Exposure time: 7 d
Method: OECD Test Guideline 221

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Remarks: Based on data from similar materials

Toxicity to fish (Chronic toxicity) : see user defined free text (Oncorhynchus mykiss (rainbow trout)): 2.9 mg/l
Exposure time: 28 d
Test Type: semi-static test

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (Daphnia magna (Water flea)): 0.02 mg/l
Exposure time: 20 d
Test Type: static test

M-Factor (Chronic aquatic toxicity) : 1

Toxicity to microorganisms : NOEC (activated sludge): 560 mg/l
Exposure time: 3 h
Method: OECD Test Guideline 209
Remarks: Based on data from similar materials

copper dinitrate:

Toxicity to fish : LC50 (Pimephales promelas (fathead minnow)): 0.0384 mg/l
Exposure time: 96 h
Remarks: Based on data from similar materials

Toxicity to daphnia and other aquatic invertebrates : LC50 (Daphnia magna (Water flea)): 0.0098 mg/l
Exposure time: 48 h
Test Type: static test

LC50 (Ceriodaphnia dubia (water flea)): 0.014 mg/l
Exposure time: 48 h
Test Type: semi-static test

Toxicity to algae/aquatic plants : NOEC (Raphidocelis subcapitata (freshwater green alga)): 0.0157 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

NOEC (Macrocystis pyrifera (brown algae)): 0.0102 mg/l
Exposure time: 19 d

EC10 (Phaeodactylum tricornutum): 0.0029 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

NOEC (Phaeodactylum tricornutum): 0.0057 mg/l
Exposure time: 72 h
Method: ISO 10253

NOEC (Skeletonema costatum (marine diatom)): 0.00754 mg/l
Exposure time: 72 h
Method: ISO 10253

M-Factor (Acute aquatic toxicity) : 10

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M-Factor (Chronic aquatic toxicity) : 10

Toxicity to microorganisms : NOEC (activated sludge): 0.23 - 0.45 mg/l
Exposure time: 30 d
Test Type: Growth inhibition

NOEC (Tetrahymena pyriformis): 3.563 mg/l
Exposure time: 48 h
Test Type: Growth inhibition

EC50 (activated sludge): 0.0025 mg/l
Exposure time: 100 d
Test Type: Growth inhibition

nitric acid ...% [C ≤ 70 %]:

Toxicity to fish : LC50 (Salmo gairdneri): 4,400 mg/l
Exposure time: 96 h
Test Type: static test
Remarks: Based on data from similar materials

LC50 (Gambusia affinis (Mosquito fish)): 6,650 mg/l
Exposure time: 96 h
Test Type: static test
Remarks: Based on data from similar materials

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 490 mg/l
Exposure time: 48 h
Test Type: static test
Remarks: Based on data from similar materials

Toxicity to algae/aquatic plants : NOEC (Marine Diatom): > 419 mg/l
Exposure time: 10 d
Test Type: static test
Remarks: Based on data from similar materials

EC50 (Marine Diatom): > 1,700 mg/l
Exposure time: 10 d
Test Type: static test
Remarks: Based on data from similar materials

Toxicity to fish (Chronic toxicity) : NOEC (Pimephales promelas (fathead minnow)): 58 mg/l
Exposure time: 30 d
Test Type: flow-through test
Remarks: Based on data from similar materials

Toxicity to microorganisms : EC50 (activated sludge): > 1,000 mg/l
Exposure time: 3 h
Method: OECD Test Guideline 209
Remarks: Based on data from similar materials

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Persistence and degradability

Components:

urea:

Biodegradability : Result: Readily biodegradable.
Biodegradation: 90 - 100 %
Exposure time: 21 d

Bioaccumulative potential

Components:

urea:

Partition coefficient: n-octanol/water : log Pow: -1.73

Mobility in soil

No data available

Other adverse effects

Product:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Toxic to aquatic life with long lasting effects.

Hygienic standards:

(Allowable concentration in air, water, including fishery waters, soil)

Components	Air	Water	Soil	Data Source
ammonium nitrate 6484-52-2	MPC - average: 0.3 mg/m ³ Limiting health hazard indicator: resorptive Hazard class: Class 4 - low hazard	MPC: 40 Milligrams per cubed decimeter Limiting health hazard indicator: toxic Hazard class: 4e MPC: 9 Milligrams per cubed decimeter (nitrite nitrogen) Limiting health hazard indicator: toxic Hazard class: 4e MPC: 0.5 Milligrams per cubed decimeter Limiting health hazard indicator: toxic Hazard class: 4	No data available	List 1 List 5

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		MPC: 0.4 Milligrams per cubed decimeter (Nitrogen) Limiting health hazard indicator: toxic Hazard class: 4 MPC: 2.9 Milligrams per cubed decimeter Limiting health hazard indicator: toxic Hazard class: 4		
urea 57-13-6	MPC - average: 0.2 mg/m ³ Limiting health hazard indicator: resorptive Hazard class: Class 4 - low hazard	MPC: 80 Milligrams per cubed decimeter Limiting health hazard indicator: toxic Hazard class: 4 Limiting health hazard indicator: general sanitary Hazard class: Class 4 - low hazard	No data available	List 1 List 4 List 5
magnesium nitrate 10377-60-3	No data available	MPC: 40 Milligrams per cubed decimeter Limiting health hazard indicator: toxic Hazard class: 4e MPC: 9 Milligrams per cubed decimeter (nitrite nitrogen) Limiting health hazard indicator: toxic Hazard class: 4e MPC: 40 Milligrams per cubed decimeter Limiting health hazard indicator: sanitary and toxicological effects Hazard class: 4 MPC: 940 Milligrams per cubed decimeter	No data available	List 5

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		Limiting health hazard indicator: toxic Hazard class: 4		
manganese dinitrate 10377-66-9	MPC - maximum: 0.01 mg/m ³ (manganese (IV) oxide) Limiting health hazard indicator: resorptive Hazard class: Class 2 - highly dangerous MPC - average: 0.001 mg/m ³ (manganese (IV) oxide) Limiting health hazard indicator: resorptive Hazard class: Class 2 - highly dangerous MPC - average chronic: 0.00005 mg/m ³ (manganese (IV) oxide) Limiting health hazard indicator: resorptive Hazard class: Class 2 - highly dangerous	MPC: 40 Milligrams per cubed decimeter Limiting health hazard indicator: toxic Hazard class: 4e MPC: 9 Milligrams per cubed decimeter (nitrite nitrogen) Limiting health hazard indicator: toxic Hazard class: 4e MAC: 45 mg/l (NO ₃) Limiting health hazard indicator: sanitary-toxicological Hazard class: Class 3 - moderately dangerous	MPC: 130 mg/kg (NO ₃) Limiting health hazard indicator: Water-migration	List 1 List 4 List 5 List 7
copper dinitrate 3251-23-8	No data available	MPC: 40 Milligrams per cubed decimeter Limiting health hazard indicator: toxic Hazard class: 4e MPC: 9 Milligrams per cubed decimeter (nitrite nitrogen) Limiting health hazard indicator: toxic Hazard class: 4e MPC: 0.001 Milligrams per cubed decimeter Limiting health hazard indicator:	MPC: 130 mg/kg (NO ₃) Limiting health hazard indicator: Water-migration	List 4 List 5 List 7

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		toxic Hazard class: 3 MPC: 0.005 Milligrams per cubed decime- ter Limiting health hazard indicator: toxic Hazard class: 3 MAC: 45 mg/l (NO3) Limiting health hazard indicator: sanitary- toxicological Hazard class: Class 3 - moderately dangerous		
nitric acid ...% [C ≤ 70 %] 7697-37-2	MPC - average: 0.15 mg/m3 (Nitric acid) Limiting health haz- ard indicator: Reflec- tory-resorptive Hazard class: Class 2 - highly dangerous MPC - maximum: 0.4 mg/m3 (Nitric acid) Limiting health haz- ard indicator: Reflec- tory-resorptive Hazard class: Class 2 - highly dangerous MPC - average chronic: 0.04 mg/m3 (Nitric acid) Limiting health haz- ard indicator: Reflec- tory-resorptive Hazard class: Class 2 - highly dangerous	No data available	No data avail- able	List 1

For explanation of abbreviations see section 16.

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13. DISPOSAL CONSIDERATIONS

Disposal methods

- Waste from residues : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.
- Contaminated packaging : Empty remaining contents.
Dispose of as unused product.
Do not re-use empty containers.
Do not burn, or use a cutting torch on, the empty drum.

14. TRANSPORT INFORMATION

ADR

- UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(manganese dinitrate, Copper dinitrate)
- Class : 9
Packing group : III
Labels : 9
Hazard Identification Number : 90
Tunnel restriction code : (-)
Environmentally hazardous : yes

UNRTDG

- UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(manganese dinitrate, Copper dinitrate)
- Class : 9
Packing group : III
Labels : 9
Environmentally hazardous : yes

IATA-DGR

- UN/ID No. : UN 3082
Proper shipping name : Environmentally hazardous substance, liquid, n.o.s.
(manganese dinitrate, Copper dinitrate)
- Class : 9
Packing group : III
Labels : Miscellaneous
Packing instruction (cargo aircraft) : 964
Packing instruction (passenger aircraft) : 964
Environmentally hazardous : yes

IMDG-Code

- UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID,

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	N.O.S. (manganese dinitrate, Copper dinitrate)
Class	: 9
Packing group	: III
Labels	: 9
EmS Code	: F-A, S-F
Marine pollutant	: yes

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

The components of this product are reported in the following inventories:

TCSI	: Not in compliance with the inventory
TSCA	: Product contains substance(s) not listed on TSCA inventory.
AIIC	: Not in compliance with the inventory
DSL	: This product contains chemical substance(s) exempt from CEPA DSL Inventory requirements. It is regulated as a pesticide subject to Pest Control Products Act (PCPA) requirements. Read the PCPA label, authorized under the Pest Control Products Act, prior to using or handling this pest control product.
ENCS	: Not in compliance with the inventory
ISHL	: Not in compliance with the inventory
KECI	: Not in compliance with the inventory
PICCS	: Not in compliance with the inventory
IECSC	: Not in compliance with the inventory
NZIoC	: Not in compliance with the inventory
TECI	: Not in compliance with the inventory

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16. OTHER INFORMATION

Full text of H-Statements

H272	May intensify fire; oxidizer.
H290	May be corrosive to metals.
H302	Harmful if swallowed.
H303	May be harmful if swallowed.
H314	Causes severe skin burns and eye damage.
H318	Causes serious eye damage.
H319	Causes serious eye irritation.
H331	Toxic if inhaled.
H341	Suspected of causing genetic defects.
H373	May cause damage to organs through prolonged or repeated exposure.
H400	Very toxic to aquatic life.
H402	Harmful to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.

Full text of other abbreviations

Acute Tox.	: Acute toxicity
Aquatic Acute	: Short-term (acute) aquatic hazard
Aquatic Chronic	: Long-term (chronic) aquatic hazard
Eye Dam.	: Serious eye damage
Eye Irrit.	: Eye irritation
Met. Corr.	: Corrosive to metals
Muta.	: Germ cell mutagenicity
Ox. Liq.	: Oxidizing liquids
Ox. Sol.	: Oxidizing solids
Skin Corr.	: Skin corrosion
STOT RE	: Specific target organ toxicity - repeated exposure
List 1	: SanPiN 1.2.3685-21 Table 1.1, Table 1.10, & Table 1.11 Maximum permissible concentration (MPC) in the air of urban and rural settlements
List 4	: SanPiN 1.2.3685-21 Table 3.13, Table 3.15, Table 3.16 & Table 3.17 Maximum permissible concentrations (MPC) of chemicals in the water of drinking systems of centralized, including hot, and non-centralized water supply, water of underground and surface water bodies of domestic drinking and cultural and domestic water use, water of swimming pools, water parks
List 5	: Order of the Russian Federal Fisheries Agency "Standards of maximum permissible concentrations of harmful substances in fishery water bodies"
List 7	: SanPiN 1.2.3685-21 Table 4.1, Table 4.2, Table 4.7, Table 4.8, Table 4.9 & Table 4.10 Maximum allowable concentration (MPC) and approximate allowable concentration (APC) of chemicals in the soil

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan);

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ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative

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